

Data Curation and Management Plans

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Lecture Note 1

Outline

- 1 Overview of Data
- 2 Data Curation
- 3 Data Quality
- 4 Data Improvement Plan: a.k.a DIPs

Overview of Data

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- It consist of clearly defined data types whose pattern makes them easily searchable.

Overview of Data

- Structured Data is data that has been catalogued into attributes and indexed for easy access.
- It has predefined attributes
- Facilitates data manipulation
- The problem with Structured Data is that the natural environment does not have a structure and there is a big effort in structuring the data.
- Such an effort often needs a lot of thought processes and manual effort. Anything that needs manual effort is not scalable.

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- The increase in the importance of Unstructured Data has given rise to the concept of Data Lakes.

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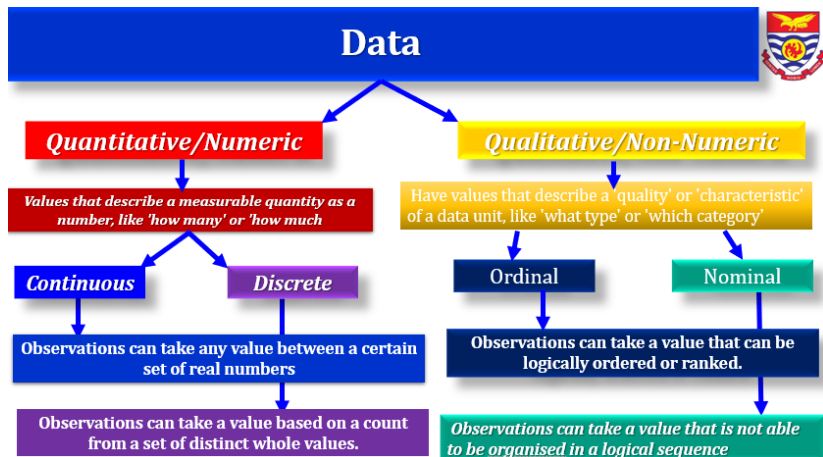
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Data Structure

Quantitative and Qualitative



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 - High Reliability and Consistency

- Data curation is the dynamic and ongoing process of data management throughout the **life cycle and usefulness of the data..**
- the process of data curation involves
 - **Preserving**: Collecting and managing and storing research data
 - **Sharing** :Making it publicly available and accessible and revealing the potential of the data across different domains.
 - **Discovering** : Enhance the re-use of the data and new combination of data

- Generally, Data curation focuses on the long-term management and preservation of data due to the high opportunity cost of losing data.
- Through the process of data curation:
 - Data is cleaned, organized, enhanced, described and preserved for public use
 - Just like old paintings, or rare books are made accessible to the public now and in the future, data curation processes ensures that data is always made available to the public
- Though modern web seem to be providing increasing methods of Data Storage, without data curation, it would be difficult to find and re-use and interpret it

Process of Data Curation: Start repository data curation service

- This should have its own staff and business models that would sustain it. Every data curation service should clearly defined the following:
- **Clearly stated Mission**
 - The commitment of the organisation
- **Clearly stated Policies**
 - State the scope of what data will be curated and establish written policies
- **Clearly Stated Targeted Audience**
 - Perform a market analysis to better understand the target audience needs and motivations for using the curation service.
 - e.g. user needs, assessment, gap analysis of existing services, stakeholder analysis.
- **cost**
 - understanding and planning for the costs involved with staffing,

Process of Data Curation: Start repository data curation service

- staffing
 - Define and allocate the organizational infrastructure to provide the data curation services at the level appropriate for your organization.
- Technological infrastructure
 - Develop (or secure from a third party) the needed repository infrastructure to securely house and store the data.

Process of Data Curation: Data Reception

- The curation should be able to provide avenues to enable data curators to submit their data for deposit into the repository.
- Recruit data for your service
- Negotiate deposit (terms of deposit)
- Obtain Author Deposit Agreement (copy rights issues)
- Facilitate transfer of the data
- Obtain the metadata and documentation
- Receive notification of data.

Process of Data Curation

- Appraise and Select
 - Data curation service should include review of submissions to allow for the selection and rejection of data that does not meet collection policy and mitigate known risk inherent to digital data.
- Processing and treatment actions for data
 - Data curation services should include processing actions for all data deposited in order to best arrange, transform, and prepare the data according to established procedures.
- Ingest and Store Data in your repository

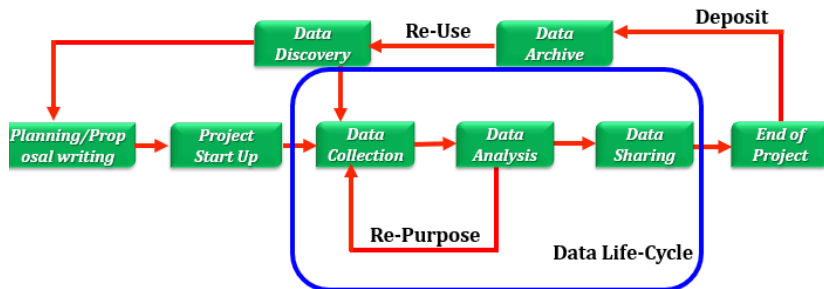
the data in a secured location using the appropriate repository infrastructure.
- Provide a descriptive metadata
 - should apply appropriate descriptive metadata and enhance existing metadata that best facilitates discovery.
- The repository should be made accessible to all and easily discoverable.

FIRST ASSIGNMENT

- (A.) Produce two pages commentary on Data Policy of your respective organizations covering the following areas:
 - 1 Structure of Data Dissemination
 - 2 Format of Data Dissemination
 - 3 Data Sharing Protocols and Procedures if any
 - 4 Data Visibility (Website/social media handles)
- (B.) Suggest ways by which the mentioned areas of data curation can be improved/implemented in your organization.

Data Practices

Creating and Managing Research Data



Best Practices for Creating Data

- Use Consistent Data Organization
- Use Standardized Naming, codes and formats
- Assign Descriptive File Names
- Perform Basic Quality Assurance/Quality Control
- Preserve Information – Use Scripted Languages
- Define Contents of Data Files; Create Documentation
- Use Consistent, Stable and Open File Formats

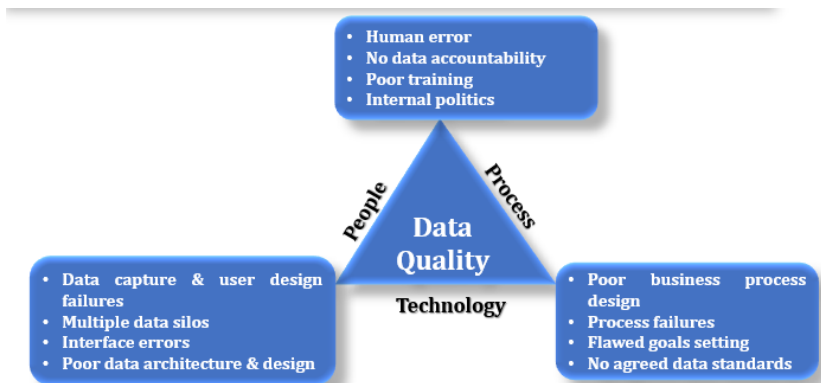
Data quality and why it matters

- Data is said to be of quality when is fit for purpose.
- That is, when it is; Accurate, Complete in all directions of the phenomenon it describes, Reliable, Accessible and Timely.
- Poor data quality could have the following implications:
 - Economic cost and revenue losses
 - Reputation cost
 - Quality of life (health and education)

Why Does Poor Data Quality Persist?

- The data world has become more complex and diffuse.
- The world changes, and data models the world
- Not recognising that poor data quality is a business problem, not an IT problem
- People will make mistakes with data
- Conflict or absence of common data definitions metadata context
- Lack of accountability for improving data

Other Sources of Poor Data Quality



Other Sources of Poor Data Quality

- Lack of accountability and governance .
- A number of organisations do not have the structures for maintaining quality data.
- People are not assigned any responsibility in terms of data and improvement of data quality.
- Hence bad data never get systematically fixed
- “If we are all supposed to be responsible, no one is responsible and nothing changes”

Making Things Better

- Implement a business led, structured Data Governance framework and organization to ensure clear priorities for data quality improvement
- Develop reusable data improvement approaches that can be applied across the organization and able to tackle traditional and new data challenges.
- Automate data quality improvement invest in a data quality toolkit to support reusable approaches .
- Link data quality to Data Architecture design improvements into data design and implementation
- Build a business case for every data quality improvement initiative so that the focus is on highest value work
- Manage each initiative via a Data Improvement Plan

Applying a Structured Data Governance Framework



Traditional Approaches to Data Quality

- Inspect data sources and highlight data deficiencies, omissions and duplication
- Develop data standards and common data definitions.
- Build business rules to enforce and police data standards
- Automate data cleaning and enhancement projects , using the business rules defined
- Embed the standards and rules into both batch and real time environments to keep the data clean
- Produce Data Quality key performance indicators and measures to monitor ongoing quality and track trends

The New Age of Data Quality

- Increased focus on real time data validation and improvement at the point of data creation, ingestion or use
 - Automated digitized processes and self service will fail if the data is not fit for purpose
 - Data validation and improvement must be done in real time on large data volumes and varieties
 - After the fact' data cleanse and improvement is now too late
 - Opportunity to exploit Internet of Things and Artificial Intelligence to develop self checking data quality capabilities
- Business users need more control over the creation and management of business rules
 - They need the ability to create business rules dynamically when preparing data
 - Different data users will potentially require the application of varying business rules
 - The paradigm where business rules are created and held centrally by IT is obsolete

The New Age of Data Quality

- End user self service data quality functionality is essential
 - Data preparation and formatting
 - Data parsing and cleansing
 - Data enhancement and enrichment
- Toolsets must support a wider variety of platforms and data types
 - Legacy and Big Data environments (e.g. Data Warehouses and Data Lakes)
 - Real time and batch
 - Structured / semi structured / unstructured data types via Data Profiling, Data Preparation and Metadata tagging

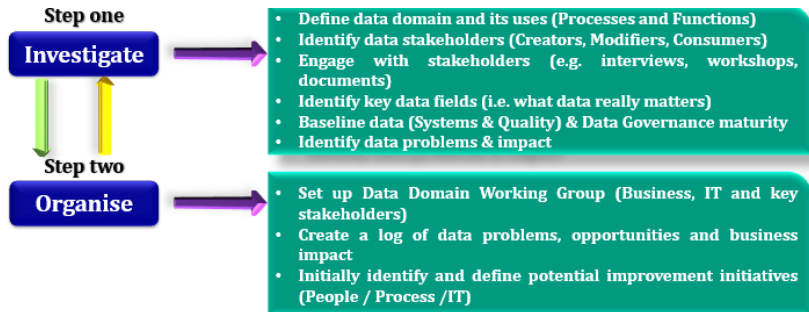
What is a Data Improvement Plan?

- A Data Improvement Plan (DIP) is a formal plan to specify and manage improvements to a specified data domain and /or data problem.

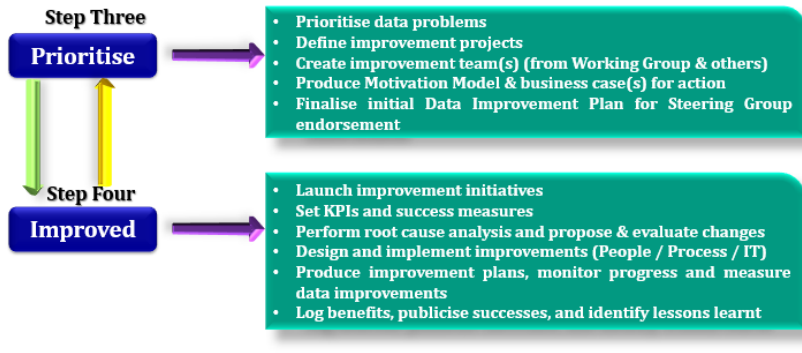
The benefits of a Data Improvement Plan are:

- sets out goals and expectations for data improvement
- acts as a focal point for all data improvement activities
- prioritises improvement activities
- can be used to track improvements and communicate successes
- can evolve to align with the changing needs of the business
- data domain DIPs can be rolled up to form the core of a company wide Data Improvement Programme

Creating a Data Improvement Plan



Creating a Data Improvement Plan



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END OF LECTURE 1